NORMAN BUI

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Education

Carleton University | *Bachelor of Computer Science (Honours),* ***3.9/4.0 GPA*** Graduated Spring 2026

- Coursework: Quality Assurance, Operating Systems, Data Structures & Algorithms, Databases, Web Development
- Awards: Dean's Honour List every year; President's, Henry Marshall Tory, and Harry S. Southam Scholarships

Experience

Canadian Tire Corporation | *Software Developer* July 2026 - Present

- Built an internal tool that reads a service's source code and writes its technical documentation automatically
- Audited a **22**-page set claim by claim against the code: **1.9%** of **1,694** claims were wrong or misleading
- Checked **3** audience-specific documents per service against rules that forbid guessing a name, a direction, or an abbreviation
- Rewrote the tool from TypeScript to Rust, with **1,100+** tests holding its output identical to the old version's

Electronic Arts (EA) | *Software Developer Intern* May 2025 - Aug 2025

- Raised automated performance test pass rates from **35% to 100%** by fixing broken logins and setup steps
- Wrote test plans for **10+** game features covering heavy traffic, long sessions, and repeat checks
- Diagnosed failing test runs, separating real game bugs from login and environment problems

Autodesk | *Software Developer Intern* Jan 2025 - Apr 2025

- Built systems that find and fix missing manufacturing data in Autodesk Fusion, with a repair step **8x** faster
- Shipped services that check and repair customer data automatically, with **100%** test coverage
- Expanded testing to cover bad input, denied access, and unusual repair cases

Kongsberg Geospatial | *Software Developer Intern* May 2024 - Aug 2024

- Built live mapping demos for aviation, defense, and government clients as the only developer on the team
- Combined several terrain and mapping data formats into one live view with real-time object tracking
- Checked that unusual terrain and mapping data loaded and displayed correctly
- Created repeatable checks for map loading, tracking, and display speed before each client demo

Health Canada | *Data Scientist Intern* Sep 2023 - Dec 2023

- Rebuilt the system that gathers chemical data, more than doubling usable sources with **100%** test coverage
- Made the department-wide chemical search more reliable, raising successful lookups by **20%**

Correctional Service Canada | *Software Developer Intern* May 2022 - Apr 2023

- Brought a public government portal up to the AA accessibility standard
- Fixed interface defects in a victim services application used by **100+** people daily
- Verified the site worked with keyboards, screen readers, and on smaller screens

Projects

Orbital Refuelling Simulator | *Python, Streamlit, scikit-learn, pytest* github.com/nbui23/orbital-refuelling

- Simulated a **410-second** spacecraft refuelling mission across **9** phases, with **9** kinds of failure to catch
- Combined fixed safety limits with machine learning that flags subtler drift, keeping the two clearly separate
- Tested all **5** parts of the system: data generation, safety rules, scoring, mission phases, and explanations

CruxOS | *Next.js, TypeScript, Prisma, SQLite, Expo, React Native* github.com/nbui23/cruxos

- Built a climbing training product: a phone app for quick logging and a web app for reviewing progress
- Turned training and recovery data into clear weekly recommendations, with the reasoning shown
- Wrote tests covering the recommendation logic, sign-in, and stored data

Skills

Languages: Python, TypeScript, JavaScript, C++, C#, Java, SQL, HTML/CSS
Frameworks & Runtimes: Node.js, Express.js, Qt, Spring Boot, Flask, Streamlit, .NET, ASP.NET
AI / ML / Data: LLMs, scikit-learn, pandas, NumPy
Databases & APIs: PostgreSQL, MySQL, SQLite, MongoDB, Firebase, Neo4j, GraphQL
Tools & Practices: Git, AWS, CI/CD, Docker, Docker Compose, Ubuntu, TDD, pytest, Mocha, Chai, Sinon, unittest, BS4, Selenium, Cucumber, PowerShell, Postman, Agile, SDLC
Spoken Languages: English, French, Vietnamese